The Identifiability of Color

A Geometric, Relativistic, and Statistical Synthesis of Metameric Projections

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November 2026

Abstract

Color is not a property of light. It is a three-dimensional linear readout of a spectral power distribution, and every readout of that kind identifies infinitely many physically distinct spectra. We treat this identification, metamerism, as a problem of identifiability: which distinctions a measurement keeps, which it discards, and which can return. Five results organize the paper. First, a linear evolution passes to the observed space exactly when it preserves the kernel of the readout; a relativistic Doppler boost acts on colorimetry as a change of observer, dilating the color-matching functions, and fails this condition, so metamers separate under motion. Second, when the evolution does not pass to the observed space, the Mori–Zwanzig projection gives the observed color an exact memory kernel, and the metameric black enters as the orthogonal term. Third, Hilbert's projective metric makes the loss precise: the boost is an isometry of the spectral cone, the colorimetric map is a contraction, and distances in the chromaticity diagram diverge at the spectral locus, where the set of metamers collapses. Fourth, identifiability under several measurements is the intersection of their kernels with the tangent structure of the physically admissible set, a quantity that changes in integer steps; statistical priors, copula dependence, active perception, and loop closure in robotic mapping are all ways of moving across those steps. Fifth, when the readouts are local, as with containers over a space of data, identification fails in two ways: the data may leave directions undetermined, or no global state may reproduce them at all, and a Farkas inequality certifies the second; one free-energy principle over the consistent set selects within the undetermined directions and contains maximum entropy, softmax attention, entropic optimal transport, and Born weights as cases. The components are classical and are cited as such; what the paper offers is a phrasing in which they are one problem. Statements are marked [Derived] when they follow from the stated setup and [Proposed] when they are hypotheses.

Typed symbol table

Symbol	Meaning	Type	First use
$\nu,\ \lambda$	frequency; wavelength	positive reals	§1
$S(\nu)$	spectral power distribution (spectrum)	nonnegative function	§1
V	space of spectra	real vector space	§1
K	cone of nonnegative spectra	convex cone in V	§1
$\bar{x}_i$	color-matching functions, $i = 1, 2, 3$	nonnegative functions	§1
C	colorimetric readout $S \mapsto (X, Y, Z)$	linear map $V \to \mathbb{R}^3$	§1
$\ker C$	metameric blacks	linear subspace of V	§1
R	Cohen projector onto $\operatorname{span}\{\bar{x}_i\}$	orthogonal projector on V	§1
(x, y)	chromaticity coordinates	point of a convex plane region	§1
$u^\mu,\ k^\mu$	observer four-velocity; photon wave vector	four-vectors	§2
$\beta,\ \gamma$	speed over c; Lorentz factor	reals	§2

Symbol	Meaning	Type	First use
D	Doppler factor	positive real	§2
η	rapidity, $\beta = \tanh \eta$	real	§2
$\mathcal{D}_\eta$	boost acting on spectra	positive linear map $V \to V$	§2
L	generator of $\mathcal{D}_\eta$	linear operator on V	§2
$T,\ \bar{T}$	evolution; its quotient on the observed space	maps $V \to V$, $W \to W$	§3
W	observed space, $C(V)$	vector space	§3
$J,\ Z_a,\ e_a$	carrier $\sum_a Z_a e_a$; amplitudes; quaternion basis	$M_2(\mathbb{C})$; complex; 2×2	§3
ϱ	positive closure $JJ^\dagger$	positive Hermitian 2×2	§3
$P,\ Q$	projector onto resolved variables; $Q = I - P$	projectors on V	§4
$X = x^\mu \sigma_\mu$	Hermitian slice element	Hermitian 2×2	§5
σ_μ	$(I_2, \sigma_1, \sigma_2, \sigma_3)$, Pauli basis	Hermitian 2×2	§5
B	Killing form of $\mathfrak{gl}(2, \mathbb{C})$	symmetric bilinear form	§5
Π	effect, $0 \le \Pi \le I_2$	Hermitian 2×2	§5
$A,\ U_\gamma$	connection; holonomy around loop γ	Lie-algebra 1-form; group element	§5
Ω	bounded convex domain	subset of $\mathbb{R}^n$	§6
$d_\Omega,\ d_K$	Hilbert metric on Ω; projective metric on K	metrics	§6
M_i	ith measurement	linear map $V \to \mathbb{R}^{m_i}$	§6
$S_{\text{phys}},\ S_{\text{adm}}$	admissible spectra; spectra consistent with data	subsets of V	§6
$\mathcal{L}_S$	lineality space of the tangent cone at S	subspace of V	§6
d_{unk}	dimension of the undetermined space	nonnegative integer	§6
$M,\ y$	stacked measurement $(M_0; \dots; M_n)$; stacked data	linear map; vector	§6
w	Farkas certificate	vector of multipliers	§6
$\mathcal{X}$	finite data space	finite set	§7
$p,\ \Delta(\mathcal{X})$	distribution on $\mathcal{X}$; probability simplex	vector; simplex	§7
$U,\ \mathbf{1}_U$	container; its indicator	subset of $\mathcal{X}$; function $\mathcal{X} \to \{0,1\}$	§7
φ	container coordinates, $\varphi(x) = (\mathbf{1}_{U_k}(x))_k$	map $\mathcal{X} \to \{0,1\}^K$	§7
$\mathcal{P}(y)$	distributions consistent with data y	polytope in $\Delta(\mathcal{X})$	§7
$\mathcal{X}_S$	points charged by some element of $\mathcal{P}(y)$	subset of $\mathcal{X}$	§7
$H(p)$	Shannon entropy, in nats	real	§7
$E,\ \tau$	cost; temperature	function $\mathcal{X} \to \mathbb{R}$; positive real	§7
q_τ	Gibbs distribution $e^{-E/\tau}/Z_\tau$	element of $\Delta(\mathcal{X})$	§7
θ	multipliers (logits)	vector in $\mathbb{R}^K$	§7
$F_i,\ \mathcal{C}$	marginal distribution functions; copula	functions into $[0,1]$	§8
π	pinhole projection	map $\mathbb{R}^3 \setminus \{Z = 0\} \to \mathbb{R}^2$	§9

The script $\mathcal{C}$ for the copula is distinct from the readout C; D is a number and $\mathcal{D}_\eta$ an operator. In (9) τ is an integration variable; elsewhere it is a temperature. In (13) p, q are points of Ω; from §7 on, p is a distribution and q_τ a Gibbs distribution.

Definitions

Definition 1 (Readout). A readout is a map from a richer structure to a smaller one. It keeps the distinctions it does not identify and discards the rest. Here the main readout is C.

Definition 2 (Metamers, metameric black). Spectra $S_1 \neq S_2$ are metamers for C when $C(S_1) = C(S_2)$, that is, when $S_1 - S_2 \in \ker C$. An element of $\ker C$ is a metameric black [4, 6].

Definition 3 (Semi-conjugacy, quotient dynamics). A map $T : V \to V$ passes to the observed space through C when there is $\bar{T} : W \to W$ with $C \circ T = \bar{T} \circ C$. Then C is a semi-conjugacy from T to $\bar{T}$.

Definition 4 (Identifiability). A state is identifiable from measurements $M_1, \ldots, M_n$ on an admissible set when no other admissible state gives the same data. It is locally identifiable when this holds in a neighborhood [38].

Definition 5 (Positive map, Hilbert metric). A linear map is positive when it sends the cone K into a cone. The Hilbert metric on a convex domain is the logarithm of a cross-ratio along chords (§6) [34].

Definition 6 (Gluing). Local data on a family of readouts glue when one admissible global state reproduces all of them. Data that agree wherever two readouts overlap can still fail to glue [49].

1 Introduction: the measurement collapse

A spectral power distribution $S(\nu)$, the power a light carries at each frequency ν, lives in an infinite-dimensional space V. Normal human vision reports three numbers. Grassmann's laws of color mixture [1] and Maxwell's matching experiments [2] established that this report is linear, and the CIE 1931 standard observer [3] fixed it as

$$C(S) = \Big(\int S(\nu)\, \bar{x}_1(\nu)\, d\nu,\ \int S(\nu)\, \bar{x}_2(\nu)\, d\nu,\ \int S(\nu)\, \bar{x}_3(\nu)\, d\nu \Big) = (X, Y, Z), \tag{1}$$

where the nonnegative color-matching functions $\bar{x}_i$ are the observer. The chromaticity $(x, y) = (X, Y)/(X + Y + Z)$ discards one more number, overall intensity.

Equation (1) has rank three, so its kernel is infinite-dimensional. Wyszecki named the elements of that kernel metameric blacks [4]: spectra that integrate to zero against every $\bar{x}_i$ and can be added to any stimulus without changing its color, within the limits set by nonnegativity. Cohen and Kappauf gave the decomposition an operator [6]: with A the matrix whose rows are the $\bar{x}_i$ on a discretized spectrum, $R = A^{\mathsf{T}}(AA^{\mathsf{T}})^{-1}A$ is the orthogonal projector onto the span of the color-matching functions, and every spectrum splits uniquely as

$$S = RS + (I - R)S, \qquad C(S) = C(RS), \qquad (I - R)S \in \ker C. \tag{2}$$

The fundamental RS carries the color; the residual $(I - R)S$ is invisible. Metamers share a fundamental and differ in their residuals.

Two consequences follow, and they frame the paper. Color is a readout, not an intrinsic property, so the question of what a color tells us is a question about which distinctions C keeps. And the residual is invisible only to this C: another observer, another illuminant, or another frame of motion supplies another readout, and the residual may stop being invisible. Metamerism is therefore an inverse problem [6, 41]. A finite set of sensory coordinates cannot reconstruct S without auxiliary constraints, and the useful questions are which constraints, which additional readouts, and which dynamics restore the lost distinctions. The thesis of the paper is that the answer has a single structure seen from five directions: the failure of a semi-conjugacy (§3), the memory kernel it forces (§4), the projective geometry that measures it (§5–§6), the local-to-global problem of gluing and selection (§7), and the statistical and active strategies that undo it (§8–§9).

2 The Doppler shift as a change of observer

The first new readout we consider is motion. A photon with wave vector k^μ has, for an observer with four-velocity u^μ, the angular frequency

$$\omega_{\mathrm{obs}} = -k_\mu u^\mu \tag{3}$$

in signature $(-,+,+,+)$ [10]. This is the Doppler shift written invariantly [9]: it acts on the light before any physiological filtering. For a source receding or approaching along the line of sight with speed βc, frequencies scale by the Doppler factor

$$D = \sqrt{\tfrac{1+\beta}{1-\beta}} = e^{\eta}, \qquad \beta = \tanh\eta, \tag{4}$$

with $\beta > 0$ for approach; for general direction $D = [\gamma(1 - \beta\cos\theta)]^{-1}$.

A frequency shift is not the whole transformation. The photon occupation number is Lorentz invariant, so specific intensity transforms with I_ν/ν^3 invariant [11]. The boost therefore acts on a spectrum as

$$(\mathcal{D}_\eta S)(\nu) = e^{3\eta} S(e^{-\eta}\nu), \qquad \mathcal{D}_{\eta_1}\mathcal{D}_{\eta_2} = \mathcal{D}_{\eta_1+\eta_2}, \tag{5}$$

a one-parameter group of positive linear maps with generator $LS = 3S - \nu\,\partial_\nu S$. (Per unit wavelength the same law reads $I_\lambda(\lambda) \mapsto D^5 I_\lambda(D\lambda)$.) A blackbody at temperature T_0 is carried to a blackbody at DT_0 [12], which serves below as a numerical check.

Substituting (5) into (1) and changing variables $\nu = e^\eta \nu'$,

$$C_i(\mathcal{D}_\eta S) = e^{4\eta} \int S(\nu')\,\bar{x}_i(e^\eta\nu')\,d\nu'. \tag{6}$$

[Derived] A boost of the source is, for colorimetry, a change of observer: the moving observer sees the original spectrum through dilated color-matching functions $\bar{x}_i(D\,\cdot)$, rescaled by D^4. This places relativistic metamer breaking in the same class as the familiar failures, change of illuminant and change of observer [7], and it is what makes the question of §3 sharp.

3 The broken commutative square

Let $T : V \to V$ be any evolution of the spectrum and $W = C(V)$ the observed space. The observed color has its own deterministic evolution exactly when the square

$$\begin{array}{ccc} V & \xrightarrow{\;T\;} & V \\ {\scriptstyle C}\downarrow & & \downarrow{\scriptstyle C} \\ W & \xrightarrow[\;\bar{T}\;]{} & W \end{array}$$

commutes for some $\bar{T}$.

Lemma 1 (Quotient dynamics). *[Derived] (i) For any surjection C and any map T, a map $\bar{T}$ with $C \circ T = \bar{T} \circ C$ exists if and only if $C(S_1) = C(S_2)$ implies $C(TS_1) = C(TS_2)$. (ii) If C and T are linear, this holds if and only if $T(\ker C) \subseteq \ker C$, and then $\bar{T}$ is unique and linear.*

Proof. (i) Define $\bar{T}(w) = C(TS)$ for any S with $C(S) = w$; the implication is exactly well-definedness, and necessity is immediate from $C(TS_i) = \bar{T}(C(S_i))$. (ii) For linear maps, $C(S_1) = C(S_2)$ means $S_1 - S_2 \in \ker C$, and $C(TS_1) = C(TS_2)$ means $T(S_1 - S_2) \in \ker C$. □

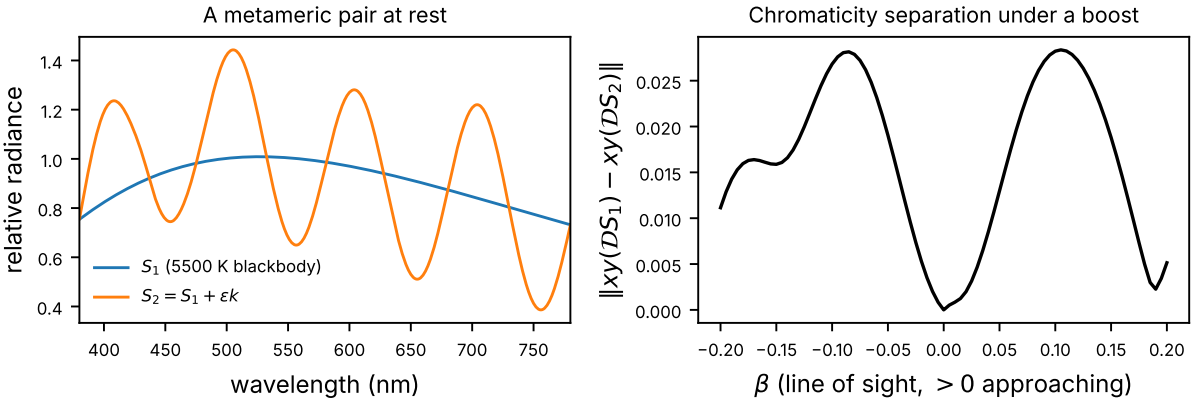


Figure 1: Left: a metameric pair, a 5500 K blackbody S_1 and $S_2 = S_1 + \epsilon k$ with $k = (I - R)g$ a metameric black. Right: chromaticity separation of the boosted pair as a function of line-of-sight speed. The separation is zero only at $\beta = 0$; it is not monotone because the dilated color-matching functions sweep across the oscillating black.

Part (i) is the lumpability condition of Kemeny and Snell for Markov chains [14]; part (ii) is the invariant-subspace condition of linear systems theory, where $\ker C$ invariant under the dynamics is the unobservable subspace [15, 16]. When the condition fails there are metamers S_1, S_2 with

$$C(S_1) = C(S_2) \quad \text{but} \quad C(TS_1) \neq C(TS_2), \tag{7}$$

and no deterministic law exists on colors alone.

Proposition 2 (Metamers separate under boosts). *[Derived]* $\mathcal{D}_\eta(\ker C) \subseteq \ker C$ *if and only if each dilated function* $\bar{x}_i(e^\eta \cdot)$ *lies in* $\operatorname{span}\{\bar{x}_1, \bar{x}_2, \bar{x}_3\}$.

Proof. By (6), $\mathcal{D}_\eta(\ker C) \subseteq \ker C$ is equivalent to $\ker C \subseteq \ker C_\eta$, where C_η has component functionals $S \mapsto \int S\, \bar{x}_i(e^\eta \cdot)$. For finitely many linear functionals f_1, f_2, f_3, g on a vector space, $\bigcap_j \ker f_j \subseteq \ker g$ holds if and only if $g \in \operatorname{span}\{f_j\}$, a standard lemma of linear algebra. □

If the color-matching functions were supported in a compact subset of $(0, \infty)$, the condition could not hold for any $\eta \neq 0$: an invariant finite-dimensional span would be invariant under every power $e^{n\eta}$, whose dilations move any nonzero function's support outside the common support of the span. Boosts therefore separate metamers. Figure 1 shows this with the analytic fit of Wyman, Sloan, and Shirley to the CIE 1931 functions [13]. The projection of the dilated functions off the original span has relative norm 0.11 at $D = 1.01$ and 0.47 at $D = 1.05$. A pair built as a 5500 K blackbody plus a metameric black from (2) matches to machine precision at rest and separates in chromaticity by 8×10^{-4} at $\beta = 0.01$ and 0.028 at $\beta = 0.1$. As a check on (5), the boosted blackbody reproduces the chromaticity of a blackbody at DT_0 to all printed digits.

The same obstruction appears in matrix form, which §5 uses. Assemble four complex amplitudes Z_a into $J = \sum_{a=0}^{3} Z_a e_a \in \mathbb{C} \otimes \mathbb{H} \cong M_2(\mathbb{C})$, with $e_0 = I_2$ and $e_j = -i\sigma_j$, and read it out by the positive closure $\varrho = JJ^\dagger$. The readout identifies every JU with J for $U \in U(2)$, and when $\det J \neq 0$ the polar decomposition $J = \varrho^{1/2} U$ shows that it identifies nothing else [17].

Example 1 (Left and right laws). [Derived] If $\dot{J} = \mathcal{L}J$, then $\dot{\varrho} = \mathcal{L}\varrho + \varrho\mathcal{L}^\dagger$, a law on ϱ alone: the square commutes. If $\dot{J} = J\mathcal{R}$, then $\dot{\varrho} = J(\mathcal{R} + \mathcal{R}^\dagger)J^\dagger = \varrho^{1/2} U(\mathcal{R} + \mathcal{R}^\dagger)U^\dagger \varrho^{1/2}$, which depends on the discarded unitary U unless the Hermitian part of $\mathcal{R}$ is a multiple of I_2. Two carriers with the same closure then have different closures later: the matrix analogue of (7).

4 Projection and memory

When the square does not commute, the observed variable still evolves, but not as a function of its present value. The Mori–Zwanzig formalism makes this exact [18, 19]. For a linear flow $\partial_\eta S = LS$, a projector P onto the resolved variables, and $Q = I - P$, splitting the flow and solving the Q part by variation of constants gives

$$\partial_\eta PS(\eta) = PLPS(\eta) + \int_0^\eta PL\, e^{(\eta-s)QLQ}\, QLPS(s)\, ds + PL\, e^{\eta QLQ}\, QS(0). \tag{8}$$

The three terms are a Markovian drift, a memory kernel, and a term driven by the initial unresolved part.

[Derived] Take $P = R$, the Cohen projector of (2), and L the boost generator of §2, so that η is rapidity along a line of sight. Then PS determines $C(S)$ and conversely, and (8) is the exact law of the observed color under a continuously varying boost. The memory kernel vanishes when $PLQ = 0$, which is the infinitesimal form of Lemma 1: the metameric blacks must not feed the fundamental. For metamers, $PS_1(0) = PS_2(0)$, so the two observed trajectories differ only through the last term of (8), the evolved metameric black. The invisible residual is not lost. It is stored in the orthogonal dynamics and returns through PLQ.

This is a general mechanism, not a property of color. Ford, Kac, and Mazur showed that a Markovian chain of oscillators induces a memory kernel on any single oscillator once the rest are eliminated [20]; the same holds whenever a readout discards variables that the dynamics couples back. Two limits of a memory kernel make the present observation sufficient. If the kernel concentrates at the present, the law forgets; if it retains the whole history without loss, the law is faithful. Every kernel between them admits no Markov closure on the observed variables. A useful general form for such kernels is the past-supported fractional integral

$$(I^\alpha f)(t) = \frac{1}{\Gamma(\alpha)} \int_{-\infty}^{t} (t-\tau)^{\alpha-1} f(\tau)\, d\tau, \tag{9}$$

which interpolates between the identity at $\alpha = 0$ and ordinary integration at $\alpha = 1$ [21].

The kernel also has a geometric reading. Path integrals of a 1-form fall into three classes [22]: exact forms depend only on endpoints; closed forms depend on the topology of the path; non-closed forms depend on the path through the curvature they enclose. A memory kernel with explicit dependence on the evaluation time lies outside all three, because extending the history changes the weight of every earlier segment. Restoring the projected-out variables as a fiber turns the kernel back into transport, which is the subject of §5.

5 The Hermitian cone, perceived color, and holonomy

The projection of §1 lands in a cone. Since the $\bar{x}_i$ are nonnegative, C maps the cone K of nonnegative spectra into a convex cone in $\mathbb{R}^3$, and Schrödinger's axioms make the space of perceived colors a regular convex cone of dimension at most three [8, 24]. Resnikoff showed that if this cone is also homogeneous, there are only two possibilities: the flat cone of Helmholtz–Stiles type, and $\mathbb{R}^+ \times SL(2,\mathbb{R})/SO(2)$, whose chromatic part is the hyperbolic plane [23, 24].

The second possibility has a matrix carrier. On the Hermitian slice of $M_2(\mathbb{C})$, write $X = x^\mu \sigma_\mu$ with $\sigma_0 = I_2$. Then

$$\det X = (x^0)^2 - (x^1)^2 - (x^2)^2 - (x^3)^2, \tag{10}$$

the Minkowski form in signature $(+,-,-,-)$; positive semidefinite X form the closed future light cone, and congruence $X \mapsto GXG^\dagger$ with $G \in SL(2,\mathbb{C})$ preserves (10) and induces the proper

orthochronous Lorentz group [28]. The positive closure ϱ of §3 lies in this cone, and its trace-one section $\frac{1}{2}(I_2 + \vec{r}\cdot\vec{\sigma})$, $|\vec{r}| \le 1$, is the Bloch ball. Restricting to real symmetric matrices drops σ_2 and leaves three real coordinates, a three-dimensional future cone acted on by $SL(2,\mathbb{R})$, whose trace-one section is a disk.

[Derived] The same algebra also carries a Lie structure and a probability rule, and it is worth stating exactly how far that goes. The anti-Hermitian part of $M_2(\mathbb{C})$ is $\mathfrak{u}(2) = \mathfrak{su}(2) \oplus \mathfrak{u}(1)$. The Killing form of $\mathfrak{gl}(2,\mathbb{C})$ is

$$B(\xi,\zeta) = 4\,\mathrm{tr}(\xi\zeta) - 2\,\mathrm{tr}\,\xi\ \mathrm{tr}\,\zeta \tag{11}$$

[50]; it vanishes on the central $\mathfrak{u}(1)$ direction and equals $4\,\mathrm{tr}(\xi\zeta)$ on $\mathfrak{su}(2)$, so the Killing form of $\mathfrak{u}(2)$ is degenerate, and the trace form agrees with it, up to the factor 4, only on $\mathfrak{su}(2)$. An effect Π, $0 \le \Pi \le I_2$, assigns to a trace-one state $\hat{\varrho}$ in the Bloch ball the number $\mathrm{tr}(\hat{\varrho}\,\Pi)$, which is the Born rule. Here the algebra supplies the arena but not the law. Gleason's theorem derives the Born rule from additivity on projections only in dimension three or more [51]; in dimension two the derivation holds for additive measures on effects [52], which is an additional assumption about what a measurement is. One carrier holding the Lorentz form (10), the Bloch ball, the Killing form (11), and Born weights is therefore a statement about where these structures live, not a derivation of any of them; what a dynamics on the carrier would impose across them is the content of open question 1 in §10.

Berthier identified this real slice with perceived color [25]: the space of perceived colors is the effect space of a rebit, a two-level real quantum system, whose chromatic state space is the Bloch disk. The disk coincides with Hering's opponent disk, its two chromatic coordinates playing the roles of the two opponent axes, and its natural metric is Klein's, which is Hilbert's metric on the disk (§6). Berthier, Garcin, Prencipe, and Provenzi then derived relativistic properties of perceived color, of the kind Yilmaz argued for experimentally [27], from this model and a single trichromacy axiom [26]. On this route the Lorentzian structure of color is citable, and it lives on the same carrier as (10).

[Derived] Transport on this carrier has holonomy. Let A be a connection with values in the Lie algebra of the structure group; transport around a loop γ is the path-ordered exponential

$$U_\gamma = \mathcal{P}\exp\Big(-\oint_\gamma A\Big), \tag{12}$$

which changes by conjugation under a gauge transformation, so its trace, the Wilson loop, is gauge invariant [29, 30]. For the hyperbolic disk, with curvature -1, Levi-Civita transport around a geodesic triangle rotates tangent vectors by the angular defect $\pi - (\alpha+\beta+\gamma)$, which by Gauss–Bonnet equals the enclosed area [31]. In group terms, the composition of two non-collinear boosts is a boost followed by a rotation, the Thomas–Wigner rotation [32, 33], and that rotation is this holonomy.

[Proposed] If changes of viewing condition act on chromatic states as $SL(2,\mathbb{R})$ transformations of the disk, the group that acts on the rebit state cone, then a sequence of such changes that returns to its starting condition leaves a residual rotation of the chromatic disk, a hue rotation, equal to the hyperbolic area the sequence encloses. A loop of adaptation would then leave a trace that depends on its shape, not on its duration. This is a test: present closed loops of background change and measure the residual hue shift against the enclosed area.

The physical boost of §2 and the perceptual transformations of this section are different group actions on different spaces: one on spectra before the readout, one on perceived states after it. Whether a principled relation exists between them is open, and §10 returns to it.

6 Hilbert geometry and the identifiability frontier

The previous sections located the loss; this one measures it. For a bounded convex domain Ω and distinct $p, q \in \Omega$, let the line through them meet $\partial\Omega$ at a and b, in the order a, p, q, b. Hilbert's metric is [34]

$$d_\Omega(p,q) = \tfrac{1}{2} \log \frac{|p-b|\,|q-a|}{|p-a|\,|q-b|}. \tag{13}$$

On the disk it is the Klein model of the hyperbolic plane, and it is invariant under projective maps of Ω because it is a cross-ratio. On a closed cone K, with $M(f/g) = \inf\{t : f \le tg\}$ and $m(f/g) = \sup\{t : f \ge tg\}$, the projective version is

$$d_K(f,g) = \log \frac{M(f/g)}{m(f/g)}, \tag{14}$$

which for nonnegative spectra is $\log[\sup(f/g)\,\sup(g/f)]$ and vanishes exactly on rays [35, 37].

Proposition 3 (Where distinguishability is lost). *[Derived] (i) Each boost $\mathcal{D}_\eta$ is an isometry of d_K on the cone of spectra. (ii) The readout C does not expand Hilbert distances: $d_{C(K)}(Cf, Cg) \le d_K(f,g)$, with Birkhoff–Hopf contraction ratio $\tanh(\Delta/4)$ when the image has finite projective diameter Δ. (iii) For metamers, $d_K(S_1, S_2) > 0$ while $d_{C(K)}(CS_1, CS_2) = 0$.*

Proof. (i) $\mathcal{D}_\eta$ is positive with positive inverse $\mathcal{D}_{-\eta}$, so $f \le tg$ if and only if $\mathcal{D}_\eta f \le t\,\mathcal{D}_\eta g$; hence M and m are unchanged. (ii) $f \le tg$ implies $Cf \le tCg$ in the image cone, so M cannot increase and m cannot decrease; the ratio is Birkhoff's theorem [35], extended by Hopf [36] and Bushell [37]. (iii) Metamers are distinct rays with equal images. □

The proposition separates three things that are easy to conflate. Motion does not destroy spectral distinguishability: for the pair of Figure 1, $d_K = 1.0667$ at rest and 1.0666 at $\beta = 0.1$ and 0.2, equal up to discretization. The readout C is where distinctions are lost, and a boost changes which ones by changing the effective observer. Because (13) is a cross-ratio, the projective metric on the image cone equals twice Hilbert's metric (13) on any affine section, and so on the chromaticity diagram, whose closure is convex: the convex hull of the spectral locus and the purple line [7].

The boundary carries the identifiability content. As q approaches the spectral locus, (13) diverges, so small changes in measurement near saturated colors are large changes in distinguishability. The metamer set behaves the same way: on the strictly convex part of the locus, the only lights with a given chromaticity are monochromatic at one wavelength, so the admissible fiber collapses to a ray. Stiles and Wyszecki found that the number of metameric object colors peaks at the achromatic point and falls toward saturation [5]. [Proposed] The size of the admissible fiber $C^{-1}(y) \cap K$ is a decreasing function of the Hilbert distance from y to the white point; the divergence of d_Ω at the locus and the collapse of the metamer set are the same phenomenon seen from the image and from the fiber.

Several measurements. Let $M_0, \dots, M_n$ be linear measurements, for instance C under different illuminants or observers. Data y_i restrict the spectrum to

$$S_{\text{adm}} = \bigcap_{i=0}^{n} M_i^{-1}(y_i) \cap S_{\text{phys}}, \qquad \ker \begin{pmatrix} M_0 \\ \vdots \\ M_n \end{pmatrix} = \bigcap_{i=0}^{n} \ker M_i, \tag{15}$$

where S_{phys} encodes nonnegativity, normalization, and band limits. At a point S of $S_{\text{phys}} = K$ that vanishes on a set Z, the admissible directions form the cone $\{\delta S \geq 0 \text{ on } Z\}$, whose largest subspace is the lineality space $\mathcal{L}_S = \{\delta S = 0 \text{ on } Z\}$. Define

$$d_{\text{unk}}(S) = \dim\Big(\bigcap_{i=0}^{n} \ker M_i \cap \mathcal{L}_S\Big). \tag{16}$$

[Derived] In a finite discretization, d_{unk} is an integer that cannot increase when a measurement is added, decreases by the number of independent new constraints, and drops when S moves onto a face of the cone. At a regular point, local identifiability holds exactly when $d_{\text{unk}} = 0$, which is Rothenberg's rank condition [38] written for linear readouts. The identifiability frontier is the set of conditions at which d_{unk} steps. Because the lineality space is used, d_{unk} counts directions undetermined in both senses; one-sided directions at the boundary are constrained by inequality, which is where the Hilbert divergence above enters.

Inconsistency. d_{unk} measures one way identification fails: too many spectra fit the data. The other is that none does. Measurements taken separately, under different illuminants, by different observers, or with noise, can leave S_{adm} in (15) empty. Write M for the stacked measurement in (15) and y for the stacked data, with normalization and band limits included as rows, so that $S_{\text{phys}} = K$. [Derived] In a finite discretization K is a polyhedral cone and (15) is a linear feasibility problem, so Farkas' lemma [53] gives an exact alternative: either $S_{\text{adm}} \neq \emptyset$, or there is a vector w with

$$\langle w, MS\rangle \geq 0 \ \text{ for every } S \in K \qquad \text{and} \qquad \langle w, y\rangle < 0. \tag{17}$$

The second alternative is an inequality on data alone, satisfied by everything a physical spectrum can produce and violated by y, and it can be checked without reconstructing S. For a single readout it is a supporting hyperplane of the image cone $C(K)$: a reported color whose chromaticity lies outside the spectral locus and purple line is certified inconsistent by a line that separates it from the diagram. With several readouts the condition is stronger than each one separately: every y_i can lie in its own image cone $M_i(K)$ while the stacked y lies outside $M(K)$. That is a failure to glue in the sense of Definition 6, and §7 treats it directly. Identifiability therefore has two frontiers: d_{unk} steps on one side, and on the other the consistent set empties. Near the spectral locus, where the fiber is already small, little noise is needed to cross from the first to the second.

7 Local readouts: gluing, identifiability, and selection

The same structure answers a question that arises whenever data come in pieces: given a space of data and a set of containers over it, how should the data be modeled? Every ingredient below is classical: Farkas' lemma [53], Jaynes' maximum entropy [54], Csiszár's I-projection [55], Vorob'ev's extension theorem [56], and Sinkhorn scaling [57, 58]. What this section adds is a phrasing in which the container question and the identifiability problem of §6 are one problem, so that gluing, identifiability, and selection become three questions about a single linear readout.

Containers as readouts. Let $\mathcal{X}$ be a finite data space and p a probability distribution on it. A container $U \subseteq \mathcal{X}$ is the readout $p \mapsto p(U) = \langle \mathbf{1}_U, p\rangle$. A marginal table on a block of variables is a family of containers, one cylinder set per cell. Stacking the normalization row $\mathbf{1}_{\mathcal{X}}$ and the indicators of containers $U_1, \ldots, U_K$ gives an integer matrix M and data y, and the consistent set is

$$\mathcal{P}(y) = \{p \in \Delta(\mathcal{X}) : Mp = y\}. \tag{18}$$

This is (15) with $S_{\text{phys}} = \Delta(\mathcal{X})$. Each point x has container coordinates $\varphi(x)$, the vector of its indicators $\mathbf{1}_{U_k}(x)$, and the data that some distribution reproduces are exactly the points of the polytope $\operatorname{conv}\varphi(\mathcal{X})$, with y's leading entry equal to one [59]. Three questions follow. Gluing: is $\mathcal{P}(y)$ nonempty? Identifiability: how large is it, as measured by d_{unk} in (16) with $\mathcal{L}_p = \{\delta : \delta_x = 0 \text{ where } p_x = 0\}$? Selection: which element should be reported?

Gluing. Data can fail to glue even when every pair of containers agrees on its overlap. Take $\mathcal{X} = \{0,1\}^3$ with coordinates x_1, x_2, x_3 and the pairwise tables on $\{x_1, x_2\}$, $\{x_2, x_3\}$, $\{x_1, x_3\}$, each saying that the two coordinates differ: mass $\frac{1}{2}$ on 01 and on 10. Every single-coordinate margin is $\frac{1}{2}$, so the tables agree on every overlap. [Derived] No distribution on $\{0,1\}^3$ reproduces them. The Farkas certificate (17) for this cover, computed by exact linear programming, is equivalent to

$$P(x_1 \neq x_2) + P(x_2 \neq x_3) + P(x_1 \neq x_3) \leq 2, \tag{19}$$

which every distribution satisfies because each assignment of three bits has zero or two unequal pairs, while the tables give 3. This is the simplest instance of contextuality in the sheaf-theoretic sense of Abramsky and Brandenburger [49], local sections compatible on overlaps with no global section, and it is Specker's triangle [60]. Whether agreement on overlaps suffices depends only on the shape of the cover. Every family of marginals that agree on overlaps glues if and only if the cover is acyclic, that is, reducible to nothing by repeatedly deleting variables that occur in only one block and blocks contained in another [56, 61]. The triangle is the smallest cyclic cover.

Selection. Among the elements of $\mathcal{P}(y)$ one has to be reported. Let $E : \mathcal{X} \to \mathbb{R}$ be a cost and $\tau > 0$ a temperature, and select

$$p^* = \operatorname*{arg\,min}_{p \in \mathcal{P}(y)} \langle E, p\rangle - \tau H(p), \qquad H(p) = -\sum_x p(x) \log p(x). \tag{20}$$

With the Gibbs distribution $q_\tau(x) = e^{-E(x)/\tau}/Z_\tau$, the objective equals $\tau \operatorname{KL}(p \,\|\, q_\tau) - \tau \log Z_\tau$, so p^* is the I-projection of q_τ onto $\mathcal{P}(y)$ [55].

Proposition 4 (Selection). *[Derived] Let $\mathcal{P}(y)$ be nonempty and let $\mathcal{X}_S$ be the set of points charged by some element of $\mathcal{P}(y)$. The minimizer p^* of (20) is unique, positive exactly on $\mathcal{X}_S$, and of the form*

$$p^*(x) \propto \exp(-E(x)/\tau + \theta \cdot \varphi(x)), \qquad x \in \mathcal{X}_S, \tag{21}$$

for some $\theta \in \mathbb{R}^K$. Equivalently, $E + \tau \log p^*$ is orthogonal to $\ker M \cap \mathcal{L}_{p^*}$. As $\tau \to \infty$, p^* tends to the element of $\mathcal{P}(y)$ of largest entropy; as $\tau \to 0$, to the minimizer of $\langle E, p\rangle$ on $\mathcal{P}(y)$ of largest entropy.

Proof. The objective is strictly convex on the compact convex set $\mathcal{P}(y)$, so the minimizer is unique. Averaging finitely many elements that charge the points of $\mathcal{X}_S$ gives $\bar{p} \in \mathcal{P}(y)$ positive on $\mathcal{X}_S$; if p^* vanished at a point of $\mathcal{X}_S$, the one-sided derivative of the objective from p^* toward $\bar{p}$ would be $-\infty$. So every $\delta \in \ker M \cap \mathcal{L}_{p^*}$ is a two-sided feasible direction at p^*, and stationarity gives $\langle E + \tau \log p^*, \delta\rangle = 0$; the constant from differentiating H drops because the normalization row is in M. Orthogonality to the kernel of M restricted to $\mathcal{X}_S$ places $E/\tau + \log p^*$ in its row space there, which is (21). For the limits, let p_τ be the minimizer and $\hat{p}$ a competitor: the maximum-entropy element of $\mathcal{P}(y)$ as $\tau \to \infty$, or any minimizer of $\langle E, p\rangle$ as $\tau \to 0$. Comparing objectives gives $H(p_\tau) \geq H(\hat{p}) - \langle E, \hat{p} - p_\tau\rangle/\tau$ in the first case and $H(p_\tau) \geq H(\hat{p})$ in the second, and every limit point of p_τ lies in the relevant set; continuity of H and strict concavity give the stated limits. □

Cost E	Containers	Selected p^*	Known as
0	any	exponential family in φ	maximum entropy [54]
$-s$	normalization only	softmax(s/τ)	Gibbs; softmax attention [62]
$-\tau \log \lvert A\rvert^2$	normalization only	$\lvert A(x)\rvert^2 / \sum_{x'} \lvert A(x')\rvert^2$	Born weights
$c(x_1, x_2)$ on $\mathcal{X}_1 \times \mathcal{X}_2$	both margins	entropic coupling, by Sinkhorn scaling	entropic transport [58]
any, $\tau \to \infty$	any	largest-entropy element of $\mathcal{P}(y)$	maximum entropy
any, $\tau \to 0$	any	largest-entropy minimizer of $\langle E, p\rangle$	linear program; argmax; transport [63]

Table 1: Cases of the selection rule (20).

$\mathcal{X}$	Containers (masses)	Glues	d_{unk}	Invisible direction; selected p^*
$\{a, b, c\}$	$\{a, b\}$: 0.7, $\{b, c\}$: 0.6	yes	0	none; $(0.4, 0.3, 0.3)$
$\{a, b, c\}$	$\{a, b\}$: 0.3, $\{b, c\}$: 0.4	no: $m_1 + m_2 \geq 1$ fails	–	–
$\{a, b, c, d\}$	$\{a, b, c\}$: 0.8, $\{b, c, d\}$: 0.7	yes	1	$b - c$; $(0.3, 0.25, 0.25, 0.2)$
$\{a, b, c, d\}$	$\{a, b, c\}$: 0.6, $\{b, c, d\}$: 0.4	yes	0	none, on a face; $(0.6, 0, 0, 0.4)$
$\{0, 1\}^2$	$\{x_1 = 0\}$: 0.6, $\{x_2 = 0\}$: 0.3	yes	1	$00 - 01 - 10 + 11$; $(0.18, 0.42, 0.12, 0.28)$
$\{0, 1\}^3$	tables on $\{x_1, x_2\}, \{x_2, x_3\}, \{x_1, x_3\}$: "differ"	no: (19)	–	–
$\{0, 1\}^3$	same three blocks, from q	yes	1	parity $(-1)^{x_1+x_2+x_3}$; $q - p^* = 0.0285$ times parity
$\{0, 1\}^3$	blocks $\{x_1, x_2\}, \{x_2, x_3\}$, from q	yes	2	x_1–x_3 interaction at each x_2; $p(x_1, x_2)p(x_2, x_3)/p(x_2)$

Table 2: Gluing, undetermined dimension, and maximum-entropy selection for small covers. In the first four rows m_1, m_2 are the masses of the two containers.

Cost and entropy therefore choose only inside the undetermined directions. Gluing and d_{unk} do not depend on E or τ, and when $d_{\text{unk}} = 0$ the containers decide everything and neither the cost nor the temperature has any effect. Table 1 lists the cases of (20) that the rest of the paper uses. The Born row is exact: with $E = -\tau \log \lvert A\rvert^2$, $e^{-E/\tau} = \lvert A\rvert^2$ at every temperature. It gives Born weights only; the Born rule also needs the measurement assumption discussed in §5.

Worked cases. Table 2 computes gluing, d_{unk}, and the maximum-entropy selection ($E = 0$) for small covers. Gluing, supports, and kernels were computed in exact rational arithmetic; the selection was computed by Newton's method on the dual of (20) and checked to machine precision against $Mp^* = y$ and the orthogonality condition of Proposition 4. In the triangle and path rows the tables come from the distribution $q = (0.20, 0.05, 0.10, 0.15, 0.05, 0.15, 0.10, 0.20)$ on $000, 001, \ldots, 111$.

Three features of Table 2 carry back to color. In the third row b and c have the same container coordinates, so they are metamers in the sense of Definition 2, and maximum entropy splits their mass evenly. In the fourth, $m_1 + m_2 = 1$ forces $p_b = p_c = 0$, $\mathcal{L}_p$ removes the direction $b - c$, and d_{unk} drops to zero: the discrete form of the collapse of the metamer set at the spectral locus (§6). In the fifth, no two points share container coordinates, yet $d_{\text{unk}} = 1$: metamerism is a property of the

kernel of the readout, not of coincidences between points. The invisible direction there is the log odds ratio, which returns in §8. For acyclic covers the maximum-entropy selection has the closed form $\prod_C p_C / \prod_S p_S$ over blocks and separators [64], here $p(x_1, x_2)p(x_2, x_3)/p(x_2)$, which makes x_1 and x_3 conditionally independent given x_2. On the cyclic triangle there is no closed form; the selection is computed iteratively [65], and it differs from the true q only along the parity direction, the three-way interaction that no pairwise container sees.

Which containers to add. [Derived] A new container U lowers d_{unk} if and only if $\mathbf{1}_U$ is not orthogonal to $\ker M \cap \mathcal{L}_p$, and then by exactly one, so reaching $d_{\text{unk}} = 0$ takes exactly d_{unk} independent containers. The number is forced; only the choice is free. Write the consistent states near p as $p + \delta$ with $\delta \in \ker M \cap \mathcal{L}_p$, and let $P_{\ker}$ be the orthogonal projector onto that space. The new datum fixes $\langle \mathbf{1}_U, \delta \rangle = \langle P_{\ker} \mathbf{1}_U, \delta \rangle$, so an error σ in the mass of U moves the recovered state by $\sigma / \|P_{\ker} \mathbf{1}_U\|$ along $P_{\ker} \mathbf{1}_U$. For $d_{\text{unk}} = 1$ with unit kernel vector k, the largest sensitivity $\max_U |\langle \mathbf{1}_U, k \rangle|$ is the sum of the positive entries of k, attained by $U = \{k > 0\}$. In Table 2 this selects $\{b\}$ in the metamer case, the agreement event $\{x_1 = x_2\}$ in the two-margin case (sensitivity 1, against $\frac{1}{2}$ for any single cell), and the even-parity set on the triangle (sensitivity $\sqrt{2}$). Two aims pull against each other. Sensitivity favors containers aligned with the kernel; the gluing guarantee favors acyclic covers. Adding $\{x_1, x_3\}$ to the path lowers d_{unk} from 2 to 1 and makes the cover cyclic. For covers made only of variable blocks, $d_{\text{unk}} = 0$ at an interior point requires a block that contains every variable, since the cells of a block span exactly the functions of that block's variables; a single event container such as the even-parity set closes the gap the triangle leaves but cannot repair its obstruction (19). For spectra the rule is the same with $\mathbf{1}_U$ replaced by the rows of the new readout: a second illuminant resolves metameric blacks in proportion to the projection of its color-matching rows onto $\ker C \cap \mathcal{L}_S$.

8 Statistical recovery of the hidden spectrum

A single spectrum cannot be recovered from data that leave $d_{\text{unk}} > 0$; no statistical method undoes an exact kernel. Statistics acts instead by changing S_{phys}: it replaces the full cone with the much smaller set of spectra that actually occur. Cohen found that the reflectances of the Munsell chips are well described by three basis functions [39], and Judd, MacAdam, and Wyszecki described daylight by a mean and two components [40]. With such finite-dimensional linear models, Maloney and Wandell showed that surface and illuminant descriptors can be recovered from sensor responses across a scene, up to scale, when the surface model has fewer dimensions than there are sensor classes [41]. In the language of §6, a model $S = \bar{S} + Bw$ replaces $\ker C$ by $\ker(CB)$, which is trivial when CB is injective.

Linear models describe the support of the ensemble but not its dependence structure. Sklar's theorem gives the rest [42]: any joint distribution of measurements $y_1, \ldots, y_n$ with marginals F_i can be written

$$F(y_1, \ldots, y_n) = \mathcal{C}(F_1(y_1), \ldots, F_n(y_n)), \tag{22}$$

with the copula $\mathcal{C}$ unique when the marginals are continuous. The marginals describe what each measurement sees; the copula describes how the measurements vary together across the ensemble. Mutual information between measurements is the negative entropy of their copula [43], so it depends on $\mathcal{C}$ alone. [Proposed] For a population of surfaces viewed under two illuminants, the copula of the two responses measures how much the second illuminant adds: a nearly comonotone copula means the second measurement is nearly redundant on that population, and a copula far from comonotone

marks the surfaces whose metameric blacks the second illuminant exposes. Principal-component reconstruction sees only second moments and misses this.

[Derived] When the readouts are margins, the dependence structure is exactly what they discard. For two binary measurements with fixed margins, the consistent joint distributions form a one-parameter family along $\mathbf{1}_{00}-\mathbf{1}_{01}-\mathbf{1}_{10}+\mathbf{1}_{11}$, the log-odds-ratio direction, and maximum entropy sets the odds ratio to one, which is independence (Table 2). With continuous margins, $h(y_1, y_2) = h(y_1)+h(y_2)-I(y_1; y_2)$, so the maximum-entropy coupling is the product, the independence copula. For discrete margins $\mathcal{C}$ in (22) is not unique, and the dependence parameter, not the copula, is the invariant object. Entropic optimal transport makes the remaining step explicit. With a cost $c(y_1, y_2)$, the selected coupling (Table 1) tends to the independence coupling as $\tau \to \infty$ and to an optimal transport plan as $\tau \to 0$, which for a strictly convex cost of $y_1 - y_2$ on the line, with atomless first margin, is the comonotone coupling [66]. Temperature therefore sweeps the copula from independence to comonotone, along the axis that the proposal above uses.

9 Translation to cyber-physical systems

Nothing above depends on color. A robot's sensors are readouts, and its perception problems have the same structure.

Sensor fusion as kernel intersection. A pinhole camera projects $\pi(X, Y, Z) = (X/Z, Y/Z)$, which is invariant under scaling of the scene; the radial direction lies in the kernel of its differential, and monocular vision cannot fix scale. A range sensor measures depth along that direction, so the intersection of kernels is trivial and fusion fixes scale, the robotic counterpart of adding an illuminant.

Active perception. Bajcsy defined active perception as controlling the sensor to obtain the data an inference needs [44], and Aloimonos, Weiss, and Bandyopadhyay showed that several vision problems that are ill-posed for a passive observer become well-posed for an active one [45]. In the terms of (16), moving the camera adds measurements M_i with different kernels, and parallax drives d_{unk} down; changing the lighting does the same for spectra.

Loop closure as holonomy. Simultaneous localization and mapping estimates poses and landmarks jointly [46, 47]. In monocular mapping the global similarity transformation is unobservable and scale drifts along the trajectory; Strasdat, Montiel, and Davison correct this by optimizing over Sim(3) and closing loops [48]. [Derived] Composing the estimated relative transformations around a cycle of the pose graph gives an element of Sim(3) that would be the identity for consistent estimates; its deviation is the holonomy of the estimated transport around that loop. A loop closure is a measurement of that holonomy. It does not remove the global gauge, but it constrains relative scale between distant parts of the map that were previously undetermined, a step across the identifiability frontier.

Attention as selection. [Derived] A row of softmax attention [62], $p_j \propto \exp(\mathbf{q} \cdot \mathbf{k}_j/\sqrt{d})$, is the selection (20) with cost $E(j) = -\mathbf{q} \cdot \mathbf{k}_j$, temperature $\sqrt{d}$, and normalization as the only container. Attention therefore already trades cost against entropy, and the score is the negative cost. A cost c_j attached to individual keys adds no structure, because $\text{softmax}((s - \lambda c)/\tau)$ is the softmax of a modified score, so cost and similarity cannot be told apart. Cost becomes new structure in two ways. With more containers, fixing the marginal over keys as well as over queries gives doubly

stochastic attention, entropic optimal transport computed by Sinkhorn scaling [57, 67]. With costs that couple assignments, attention weights become marginals of a graphical model [68], computed exactly by message passing when the coupling graph is acyclic [64] and approximately otherwise. Acyclicity therefore governs both whether local data glue (§7) and whether the selected model can be computed locally.

10 Synthesis and open questions

The paper has followed one structure in five forms. A readout identifies states (§1). If the evolution respects the identification, the observed variable has its own law; if not, the identification breaks and metamers separate, and a relativistic boost breaks it because it acts as a change of observer (§2–§3). The broken law is not lost but becomes a memory kernel whose source is the discarded part (§4). The cone on which observations live carries a Lorentzian and a hyperbolic geometry, transport on it has holonomy, and Hilbert's metric shows that motion preserves spectral distinguishability while the readout removes it, most severely near white and least near the spectral locus (§5–§6). When the readouts are local, the same structure decides whether pieces of data glue, how much they leave undetermined, and how a model is selected within what remains, and the selection rule contains maximum entropy, softmax, entropic transport, and Born weights as cases (§7). Recovery comes from constraining the admissible set statistically or by adding readouts actively (§8–§9).

Open questions.

1. [Proposed] *An eight-dimensional carrier.* The full algebra $\mathbb{C} \otimes \mathbb{H} \cong M_2(\mathbb{C})$ has eight real dimensions. One hypothesis assigns spacetime coordinates (X, Y, Z, T) and chromatic channels (R, G, B, W) to it, so that physical boosts and perceptual transformations act on one object. Section 5 shows that both structures live on Hermitian slices of this algebra, the physical one on the full slice and the perceptual one on its real subslice, but no result here establishes the assignment, and it is stated as a proposal.

2. *Relating the two Lorentz actions.* Is there a map from the Doppler action (5) on spectra to the $SL(2, \mathbb{R})$ action on the chromatic disk under which the readout intertwines them, at least to first order in η? Lemma 1 says exact intertwining fails, so the question is about the best approximate quotient and its memory kernel.

3. *Fiber size against Hilbert distance.* Test the proposal of §6 by computing the volume of $C^{-1}(y) \cap K$ within a reflectance model as a function of $d_\Omega(y, \text{white})$.

4. *Holonomy of adaptation.* Test the proposal of §5 with closed loops of background change.

5. *Gluing across observers.* For observers or illuminants measured separately, compute the smallest change to the data that makes (15) consistent; its dual maximizes the violation over normalized certificates of the form (17). Whether real multi-illuminant data sit near this frontier, and whether the certificate localizes to particular wavelength bands, is open.

6. [Proposed] *Self-consistent temperature.* For $p = \text{softmax}(-E/\tau)$, $dH/d\tau = \text{Var}_p(E)/\tau^3 \geq 0$. If the temperature depends on the entropy it produces, $\tau = f(H)$, the states solve $H = h(f(H))$, with $h(\tau)$ the entropy at temperature τ. [Derived] A decreasing f gives exactly one state. Two or more require f increasing with loop gain $f'(H)\,\text{Var}_p(E)/\tau^3 > 1$ somewhere, and a state is stable under relaxation when its gain is below one. For fifty options, one favored by one unit

of score, and $f(H) = \tau_0 + 0.06H$, two stable states coexist for τ_0 between 0.098 and 0.122; at $\tau_0 = 0.10$ they are $H \approx 0.03$ nats, near the argmax, and $H \approx 2.76$ nats, nearly diffuse, separated by an unstable state at $H \approx 2.11$, so a slow sweep of τ_0 up and down shows hysteresis. With feedback slope 0.04 or less there is never more than one state. The system is finite, so this is not a phase transition in the thermodynamic sense, which needs an infinite system [69]; the feedback alone produces the coexistence, the mean-field mechanism of the Curie–Weiss model [70]. The proposal is that rigid and fluid modes of selection, in perception and elsewhere, correspond to these coexisting states, and the test is hysteresis under a slow sweep of the conditions that set τ_0.

Whether the subject is relativistic metamerism or a robot's sensor suite, measurement decides which differences stay visible, dynamics decides which of the hidden ones return, and identifiability depends on the geometry of what stays invisible within the limits of what is physically possible.

Acknowledgments

I thank Alastair for posing the container question that §7 answers, and for the challenge that a known answer can still be phrased in ways that are novel and rigorous; §7 is an attempt at such a phrasing.

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